

NOT A DROP TO DRINK

A government study in Bihar finds alarming levels of arsenic in the ground water in 12 districts with the fear that the problem could spread to the other parts of the Gangetic plain



Photographs by RANJAN RAH1

DEADLY METAL

The 12 arsenic affected districts of Bihar

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| 1. Patna | 7. Darbhanga |
| 2. Buxar | 8. Begusarai |
| 3. Bhojpur | 9. Munger |
| 4. Saran | 10. Khagaria |
| 5. Vaishali | 11. Bhagalpur |
| 6. Samastipur | 12. Katihar |



■ By Amitabh Srivastava

People in 12 districts of Bihar are grappling with an unusual enemy: water. Thousands who depend on underground water from local wells or gushing from hand pumps have now been told that what they drink could in fact be poison. High levels of arsenic have been found in the underground water across the state. Arsenic is a toxin that dissolves in the bloodstream, rendering the victim susceptible to diseases of the skin, bones and also cancer of the liver, kidney, gall bladder and the intestines.

A study by Bihar's Public Health and Engineering Department (PHED)

shows that of the 38 districts in the state, as many as 12 have an average arsenic content of 500 parts per billion (ppb) against the World Health Organisation norm of 10ppb. Other than Patna, the places that have been severely affected include districts like Bhojpur, Bhagalpur, Vaishali, Katihar, Samastipur, Darbhanga, Buxar, Khagaria, Munger, Saran and Begusarai. Arsenic, an odourless and tasteless metalloid, occurs naturally in the environment and is sometimes a by-product of agricultural and industrial activities.

"Arsenic is a toxic substance. The point of concern is that its impact on one's health becomes evident only after prolonged consumption," says

LIQUID BOMB: Officials using testing kits in Maner, Patna (far left)

state PHED Minister Prem Kumar. To make matters worse, the response of the state Government has been lackadaisical towards the problem.

There are various theories that try to explain increased levels of arsenic in water, some of them being oxidation, deteriorating conditions of the aquifers and increased organic inflow. But according to experts, the rise cannot be ascribed to man-made reasons alone. There are geographical reasons behind it as well. "If you draw a line on the map of Asia along the arsenic-affected regions of Bangladesh, West Bengal, Bihar, the Gangetic plains of Uttar